

Which Divergences Control Representations?

Tail Sensitivity and the Limits of Identifiability

Anonymous Author(s)
Anonymous Institution

Abstract

Identifiability theory says that two models in the softmax family with *equal* likelihood have representations equal up to an invertible linear map. Trained models agree only approximately, and it is known that vanishing Kullback–Leibler divergence does not imply similar representations — leaving open which notions of distributional closeness, if any, do. We show the answer turns on one property. Representational error is an *unweighted* spread of log-likelihood ratios, and those ratios are largest where probability mass is smallest, whereas classical divergences weight each label by its own mass. We call the gap *tail sensitivity* and index it by Rényi order α . Exact asymptotics give a critical order $\alpha_{\text{sym}}^* > 1$, so every order $\alpha \leq 1$ — total variation, Hellinger, Jensen–Shannon and KL — fails outright; a dichotomy shows $\alpha_{\text{sym}}^* = \infty$ only for linearly equivalent pairs, and a graded bound gives similarity at rate $O(1/\alpha)$ above threshold. Two consequences are practical. KL’s guarantee is unusable for any model, since the $(M+1)$ -th largest probability is at most $1/(M+1)$; but a sharpened perturbation bound certifies 58% of retrained CIFAR-10 pairs where the previous constant certified none. We supply a four-number diagnostic, computable in one forward pass, that bounds in advance what a likelihood comparison can reveal. On 3,800 synthetic and 705 CIFAR-10 model pairs spanning four representation dimensions we show that matched likelihood is least informative exactly when models are confident and representations wide, and confirm the mechanism causally: rescaling logits leaves representational dissimilarity invariant to machine precision while raising the confidence floor nine orders of magnitude, and the association between likelihood and representation strengthens significantly (+0.109, 95% CI [+0.044, +0.202]).

1 Introduction

Whether independently trained networks learn related representations is an active question [Kornblith et al., 2019, Klabunde et al., 2025], and any answer depends on what counts as “the same” representation. Identifiability theory supplies a principled choice: in the softmax family of Eq. (1), two models with equal likelihood have embeddings equal up to an invertible linear map [Roeder et al., 2021, Khemakhem et al., 2020, Lachapelle et al., 2023], so a similarity measure should be invariant to exactly that class.

Exact equality is not what happens. Trained models agree approximately, and Nielsen et al. [2025] showed this is not enough: two models can have vanishing KL divergence and dissimilar representations. It is tempting to read that as a fact about KL and look for a better divergence. We argue it is a fact about a *property* KL lacks, and that naming the property settles the question for a whole family at once.

Tail sensitivity. Write $r_y(x) = \log p(y|x) - \log p'(y|x)$. The representational error between two models is a variance of these log-ratios across labels (Sec. 4) — an *unweighted* spread. But r_y is largest exactly where $p(y|x)$ is smallest, while every classical divergence weights label y by its own mass. A divergence can control representations only if it stays sensitive to log-ratios at labels carrying vanishing mass. The Rényi order α tunes precisely this weighting,

from the mass-weighted average at $\alpha = 1$ to the worst-case ratio at $\alpha = \infty$, so the question becomes one about position on a single axis.

Contributions.

- *Control* (Def. 1), making divergence–representation claims falsifiable and uniform over a model class.
- Exact rate asymptotics and a closed-form critical order (Thm. 3), verified to solver precision; hence all orders $\alpha \leq 1$ fail (Cor. 4).
- A dichotomy (Thm. 11), graded control (Thm. 12) and an explicit finite cap (Cor. 13), confining the controlling orders to an interval.
- A perturbation bound improving the known modulus by $\sqrt{M}$ (Thm. 8), whose non-vacuity is capped by an identity (Prop. 16) that trained networks nonetheless satisfy: it certifies 58% of retrained CIFAR-10 pairs against 0% before.
- A four-number diagnostic (Alg. 2) and experiments on 3,800 synthetic and 270 CIFAR-10 pairs, including a confound-free temperature intervention.

2 Setup

Models are pairs $(\mathbf{f}, \mathbf{g}) \in \Theta$ with embedding $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^M$ and unembedding $\mathbf{g} : \mathcal{Y} \rightarrow \mathbb{R}^M$:

$$p_{\mathbf{f}, \mathbf{g}}(y | x) = \frac{\exp(\mathbf{f}(x)^\top \mathbf{g}(y))}{\sum_{y' \in \mathcal{Y}} \exp(\mathbf{f}(x)^\top \mathbf{g}(y'))}. \quad (1)$$

Write $\mathbf{f}_0(x) = \mathbf{f}(x) - \mathbf{f}(x_0)$, $\mathbf{g}_0(y) = \mathbf{g}(y) - \mathbf{g}(y_0)$; let $\mathbf{L}$ have columns $\mathbf{g}_0(y_i)$ and $\mathbf{N}$ columns $\mathbf{f}_0(x_i)$, $i = 1..M$. Under the *diversity* condition — that both sets span $\mathbb{R}^M$ — equal likelihoods force $\mathbf{f} = \mathbf{A}\mathbf{f}'$ and $\mathbf{g}_0 = \mathbf{A}^{-\top} \mathbf{g}'_0$ for one invertible $\mathbf{A}$; write $(\mathbf{f}, \mathbf{g}) \sim_L (\mathbf{f}', \mathbf{g}')$. Diversity is mild where labels are plentiful ($|\mathcal{Y}| \sim 10^4$ against $M \sim 10^3$ in language models) but binds for image classifiers with fewer classes than representation dimensions.

Following Nielsen et al. [2025] we measure dissimilarity with $d_{\mathbf{f}, \mathbf{g}}$, built from the PLS-SVD statistic d_{SVD} applied to $\mathbf{L}^\top \mathbf{f}(x)$ and $\mathbf{N}^\top \mathbf{g}(y)$ (Alg. 3, App. D). It vanishes exactly on $\sim_L$ -equivalent pairs, and unlike CCA-based measures is not invariant to every invertible linear map: its invariance class is fixed by the identifiability result rather than chosen. We compare against m_{CCA} , linear CKA and Procrustes throughout.

Why this invariance class. The choice is not free. CCA-based measures are invariant to *every* invertible linear map, so they attain their maximum on pairs related by maps that identifiability does not license, and cannot distinguish $\sim_L$ -equivalence from a weaker linear relation; CKA restricts to orthogonal maps and isotropic scaling, which is narrower than $\sim_L$ in the other direction. $d_{\mathbf{f}, \mathbf{g}}$ is invariant to exactly the transformations that leave the model likelihood unchanged, which is what a measure must respect if “similar” is to mean “indistinguishable from the data”. It also couples embeddings and unembeddings — $\mathbf{L}$ depends on $\mathbf{g}$ and $\mathbf{N}$ on $\mathbf{f}$ — so it cannot be fooled by a pair whose embeddings match while their unembeddings do not. We do not claim this is the only defensible choice, and we report m_{CCA} , CKA and Procrustes alongside it throughout (Sec. 5); they agree within each representation dimension.

Definition 1 (Control). A divergence D controls $d_{\mathbf{f}, \mathbf{g}}$ on Θ if there is a nondecreasing h with $h(0^+) = 0$ such that for all pairs in Θ , $D(p_{\mathbf{f}, \mathbf{g}}, p_{\mathbf{f}', \mathbf{g}'}) \leq \epsilon \Rightarrow d_{\mathbf{f}, \mathbf{g}} \leq h(\epsilon)$.

Control is uniform over Θ ; a divergence may fail on Θ yet control on a subclass. Since D_α is nondecreasing in α , the set of controlling orders is upward closed, so $\alpha_\Theta^* := \inf\{\alpha : D_\alpha \text{ controls}\}$ is well posed. The rest of the paper brackets it.

3 No Order at or below One Controls

Fix x , write $s_y = \cos(\mathbf{f}(x), \mathbf{g}(y))$, $\hat{y} = \arg \max_y s_y$, and let $a_y = s_{\hat{y}} - s_y > 0$ and $b_y = s'_{\hat{y}} - s'_y > 0$ be the *margin gaps* of the two models. Along the norm-growth family of

Nielsen et al. [2025] the unembeddings have common norm ρ , so logits are $c\rho s_y$ and

$$p(y|x) = \frac{e^{-c\rho a_y}}{Z_\rho}, \quad p'(y|x) = \frac{e^{-c\rho b_y}}{Z'_\rho}, \quad (2)$$

with $a_{\hat{y}} = b_{\hat{y}} = 0$. Every off-top term vanishes as $\rho \rightarrow \infty$, so $Z_\rho, Z'_\rho \rightarrow 1$:

Lemma 2 (Co-concentration). *Both models converge to the same point mass, $p(\hat{y}|x) \rightarrow 1$ and $p'(\hat{y}|x) \rightarrow 1$, so $\|p - p'\|_{\text{TV}} \rightarrow 0$; meanwhile $|r_y| = c\rho|b_y - a_y| + O(1) = \Theta(\rho)$ whenever $a_y \neq b_y$.*

This is the mechanism in one line: the two distributions become indistinguishable in any mass-weighted sense while their log-ratios diverge. Whether a divergence notices depends on how it weights the tail. Define the *tilted exponent* $E_\alpha(y) = \alpha a_y + (1 - \alpha)b_y = b_y + \alpha(a_y - b_y)$, which governs the Rényi integrand $p^\alpha p'^{1-\alpha} \propto e^{-c\rho E_\alpha(y)}$.

Theorem 3 (Rate lemma and critical order). *Let*

$$\alpha^* = \min_{y : b_y > a_y} \frac{b_y}{b_y - a_y}. \quad (3)$$

Then $\alpha^ > 1$, and as $\rho \rightarrow \infty$: (i) for $\alpha < \alpha^*$, $D_\alpha = \Theta(e^{-c\rho\kappa(\alpha)})$, where κ is the smallest exponent in $\{E_\alpha(y)\} \cup \{a_y\} \cup \{b_y\}$ carrying a non-vanishing coefficient; (ii) at $\alpha = 1$ all first-order coefficients cancel and $d_{\text{KL}} = \Theta(\rho e^{-c\rho a_{\min}})$; (iii) for $\alpha > \alpha^*$, D_α grows linearly in ρ with slope $\max_y (-E_\alpha(y))/(\alpha - 1)$.*

Sketch. Expanding $\log \sum_y p^\alpha p'^{1-\alpha}$ to first order and grouping the three exponent families by value, the coefficient at value v is $\#\{E_\alpha = v\} - \alpha\#\{a = v\} + (\alpha - 1)\#\{b = v\}$. A label with $a_y = b_y$ has $E_\alpha(y) = a_y = b_y$ and contributes $1 - \alpha + (\alpha - 1) = 0$: such labels *cancel exactly* and must be excluded, which is the one non-obvious step. At $\alpha = 1$ every coefficient vanishes, forcing the second-order term and its polynomial prefactor. Finally E_α is affine in α with $E_0(y) = b_y > 0$, decreasing only when $a_y < b_y$, and vanishing at $b_y/(b_y - a_y)$; since $0 < b_y - a_y < b_y$, that value exceeds 1. Full proof in App. B. $\square$

That $\alpha^* > 1$ is the whole negative result:

Corollary 4. *Total variation, squared Hellinger, Jensen–Shannon and Kullback–Leibler all fail to control $d_{\mathbf{f}, \mathbf{g}}$. The known KL counterexample is the boundary case $\alpha = 1$ of a one-parameter statement.*

Verification is exact (Tab. 1): across 295 geometries the median relative error in the crossover is 0.00%, decay rates match to $< 2.3\%$ and growth slopes to $< 0.1\%$. Omitting the cancellation rule inflates predicted rates by 40–70%.

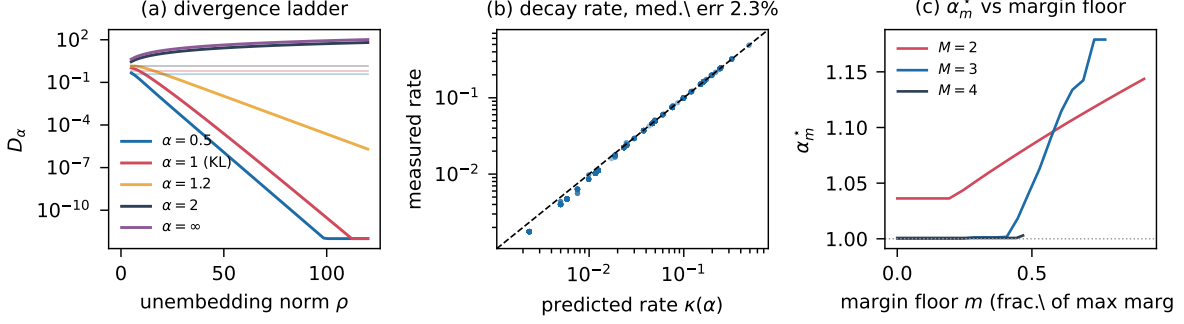


Figure 1: (a) Divergence ladder along the norm-growth family of Eq. (2). Every order $\alpha \leq 1$, KL included, decays toward zero; orders above α^* grow linearly in ρ , as Thm. 3(iii) predicts. The faint horizontal lines are $d_{f,g}$, m_{CCA} and d_{LLV}^λ : the representations do not move at all, so every decaying curve is a divergence losing sight of a difference that is still there. (b) Predicted against measured exponential decay rate over 739 (geometry, order) pairs spanning 2.3 decades; median relative error 2.3%. This is the panel that exercises the cancellation rule of Thm. 3(i) — labels with $a_y = b_y$ must be excluded, and omitting them inflates predicted rates by 40–70%. (c) α_m^* , the critical order restricted to inputs whose margin exceeds m , for three label counts. Without conditioning, α^* is pinned just above 1 by whichever single input lies nearest a decision boundary: bisectors have codimension one, so any full-dimensional input support crosses them. The threshold is a property of the worst-margin input, not of the representational geometry.

Table 1: Rate lemma against exact computation, single input, $k = 7$, $\rho \leq 160$. The lemma is per-direction and verifies in both: $\alpha_{fwd}^* = 1.349915$ (bisected 1.349919) and $\alpha_{rev}^* = 1.276277$ (bisected 1.276430), so $\alpha_{sym}^* = 1.2763$.

α	branch	predicted	measured
0.5	exp	0.2866	0.2866
1.0 (KL)	exp	0.2866	0.2866
1.1	exp	0.2047	0.2044
1.3	exp	0.0409	0.0401
1.6	linear	0.3414	0.3414
2.0	linear	0.5325	0.5324
∞	linear	0.8623	0.8623

Theorem 5 (Uniform separation). *Along the norm-growth family d_{SVD} is exactly independent of ρ , so any non- $\sim_L$ -equivalent pair stays separated by a constant $c > 0$ of the geometry alone.*

Remark 6 (Averaging preserves the rate, not the constant). Thm. 3 is per-input. Averaging over an input distribution leaves the decay exponential rather than turning it into a power law ($R^2 = 1.00$ against 0.91 for a power-law fit), but the rate is set by the *minimum* margin in the support: it falls from 0.193 to 0.018 when clusters are allowed to touch the decision boundary. Rates should therefore be quoted against a margin floor, which is what Fig. 1(c) does.

Remark 7 (Both directions). Rényi divergence is asymmetric, so the meaningful quantity is $\alpha_{sym}^* = \min\{\alpha^*(p||p'), \alpha^*(p'||p)\}$: a family is a counterexample only if both directions stay small. Computing only the forward direction reports ∞ whenever $b_y \leq a_y$ for every y , even though the reverse divergence then diverges.

4 What Does Control, and How Far

Everything positive follows from one perturbation bound. With $z_1 = \mathbf{L}^\top \mathbf{f}(x)$, $z_2 = \mathbf{L}'^\top \mathbf{f}'(x)$ and $\Delta_l = z_{1,l} - z_{2,l} = r_{y_l} - r_{y_0}$, set $sd_l = \text{std}(\Delta_l)$, $\tau_l = |s_l - s'_l|$ and $\psi_l = \min(s_l, s'_l)$, where s_l, s'_l are the coordinate standard deviations. Note $s_l = \psi_x(y_l; p)$, the non-degeneracy already present in the LLV construction.

Theorem 8 (Perturbation bound). *Let $\lambda_{\max} = \lambda_{\max}(\text{Corr}(z_1)) \leq M$. Then*

$$d_{f,g} \leq \sqrt{\lambda_{\max}} Q, \quad Q^2 = \frac{1}{M} \sum_l \left(\frac{\tau_l + sd_l}{\psi_l} \right)^2. \quad (4)$$

In particular $D_\infty \leq \epsilon$ gives $d_{f,g} \leq 4\sqrt{\lambda_{\max}} \epsilon / \psi_{\min}$.

Two steps are new relative to a naive estimate, and both are free (App. C). Hoffman–Wielandt replaces a per-singular-value Weyl bound, saving $\sqrt{M}$:

Proposition 9 (Tighter spectral step). $\sum_i (\sigma_i - \tilde{\sigma}_i)^2 \leq \|E\|_F^2$ and Cauchy–Schwarz give $d_{SVD} \leq (\sum_l \text{Var}[z'_l - w'_l])^{1/2}$, improving the known $(M \sum_l \text{Var})^{1/2}$ by $\sqrt{M}$.

Second, τ_l is kept separate from sd_l rather than bounded by it, which is wasteful because the difference of standard deviations is much smaller than the standard deviation of the difference (median τ_l/sd_l is 0.25–0.53). Jointly they tighten the bound by 1.65–2.37 $\times$, and on CIFAR-10 by 1.67 $\times$ — the difference between certifying nothing and certifying a majority (Sec. 5).

Theorem 10 (KL on Θ_δ). *Let $\mathcal{Y}_m \subseteq \mathcal{Y}$ satisfy diversity with $|\mathcal{Y}_m| \geq M + 1$, take $\mathcal{Y}_{LLV} \subseteq \mathcal{Y}_m$ and $\delta_m = \min_x \min_{y \in \mathcal{Y}_m} \min\{p, p'\}$. If $d_{KL} \leq \epsilon$ then $d_{f,g} \leq 4\sqrt{2\lambda_{\max}} \epsilon / (\delta_m \psi_{\min})$.*

Theorem 11 (Dichotomy). *Under diversity, $\alpha_{\text{sym}}^* = \infty \iff p_{\mathbf{f},\mathbf{g}} = p_{\mathbf{f}',\mathbf{g}'} \iff (\mathbf{f}, \mathbf{g}) \sim_L (\mathbf{f}', \mathbf{g}')$. Every non-equivalent pair therefore has a finite critical order.*

Sketch. Both directions infinite forces $a_y(x) = b_y(x)$ for all x, y . Writing $R_y(x) = \mathbf{f}(x)^\top \mathbf{g}(y) - \mathbf{f}'(x)^\top \mathbf{g}'(y)$ and using $a_y - b_y = R_{\hat{y}} - R_y$, this says $R_y(x) = c(x)$ is independent of y ; an additive $c(x)$ cancels in the softmax normaliser, so the conditionals coincide and linear identifiability applies. The converse is immediate. $\square$

A regression test over 329 pairs gives $\max |d_{\mathbf{f},\mathbf{g}}| = 5.0 \times 10^{-4}$ in the infinite class against $\min |d_{\mathbf{f},\mathbf{g}}| = 6.5 \times 10^{-2}$ in the finite class, the residual decaying as $O(1/n)$.

Theorem 12 (Graded control). *With $\Delta = \sup_{x,y} a_y(x)$, if $\alpha_{\text{sym}}^* \geq A$ then $d_{\mathbf{f},\mathbf{g}} \leq 4\sqrt{M}\Delta/(\psi_{\min}A) = O(1/A)$.*

Sketch of Thm. 12. $\alpha_{\text{sym}}^* \geq A$ gives $|a_y - b_y| \leq \max(a_y, b_y)/A \leq \Delta/A$; since $a_y - b_y = R_{\hat{y}} - R_y$, the map $y \mapsto R_y(x)$ lies within Δ/A of a constant, so $|\Delta_l| \leq 2\Delta/A$. Apply Thm. 8. $\square$

Corollary 13 (The ceiling is finite). *For any $d_0 > 0$, $\alpha_0(d_0) := \sup\{\alpha_{\text{sym}}^* : d_{\mathbf{f},\mathbf{g}} \geq d_0\} \leq 4\sqrt{M}\Delta/(\psi_{\min}d_0) < \infty$.*

Together: $\alpha_{\text{sym}}^* > 1$ from below, $\alpha_{\text{sym}}^* < \infty$ from above, and an explicit cap between. Searches over permutation classes, dimensions, norm profiles and angular arrangements place the empirical ceiling at 1.35 (App. F).

4.1 What can and cannot be certified

The three obstructions are not alike.

(i) Orders $\alpha \leq 1$ fail qualitatively (Cor. 4); no constant repairs them.

(ii) Thm. 10 is non-vacuous only for $\epsilon < \delta_m^2/64$, and the floor cannot be lifted:

Proposition 14 (KL’s floor is capped). *For any model $\delta_{M+1} \leq 1/(M+1)$, since $M+1$ entries each exceeding it would sum to more than one. Hence (10) is below 1 only for $\epsilon < 1/(64(M+1)^2)$; at $M=2$ this is $\epsilon < 1.7 \times 10^{-3}$.*

On CIFAR-10 the realised requirement is $\epsilon < 1.2 \times 10^{-31}$ against observed KL of order one. Quantile floors do not help: discarding the worst 10% of inputs buys nine orders and leaves thirteen. This is tail insensitivity written as an inequality — d_{KL} is a mass-weighted average of log-ratios while ρ_2 is their unweighted spread, and conversion pays the ratio of the two measures.

Corollary 15 (Confidence, not quality). *Two models can have arbitrarily small d_{KL} and arbitrarily large $d_{\mathbf{f},\mathbf{g}}$ precisely when δ_m is small. Since accurate classifiers and language models place negligible mass on most labels, these are exactly the regimes in which likelihood agreement carries least representational information — raising δ_m should*

Algorithm 1: Critical order α_{sym}^*

Input: $\log p, \log p'$ on N inputs; margin floor $m \geq 0$

Output: α_{sym}^* ; $= \infty$ iff $p = p'$ (Thm. 11)

$\hat{y} \leftarrow \arg \max_y \log p$; $\hat{y}' \leftarrow \arg \max_y \log p'$;
keep inputs with $\hat{y} = \hat{y}'$ and $\min_{y \neq \hat{y}} a_y \geq m$;

report the retained fraction;

$\alpha_{\rightarrow} \leftarrow \min\{b_y/(b_y - a_y) : b_y > a_y\}$;

$\alpha_{\leftarrow} \leftarrow \min\{a_y/(a_y - b_y) : a_y > b_y\}$;

return $\min(\alpha_{\rightarrow}, \alpha_{\leftarrow})$; // both: Rem. 7

Algorithm 2: Pre-comparison diagnostic

Input: two trained models; held-out inputs

Output: what any likelihood comparison between them can reveal

$\delta_{M+1} \leftarrow \min_x (M+1)$ -th largest $p(y|x)$;

if observed $d_{\text{KL}} > \delta_{M+1}^2/64$ **then** **KL cannot certify**;

$\lambda_{\max} \leftarrow \lambda_{\max}(\text{Corr}(z_1))$; **ceiling** $\leftarrow 1/(2\lambda_{\max})$;

$Q^2 \leftarrow \frac{1}{M} \sum_l ((\tau_l + \text{sd}_l)/\psi_l)^2$; **bound**

$\leftarrow \sqrt{\lambda_{\max} Q}$;

if bound < 1 **then certified:** $d_{\mathbf{f},\mathbf{g}} \leq \text{bound}$;

$\alpha_{\text{sym}}^* \leftarrow \text{Alg. 1}$; // order at which the pair becomes visible

therefore make the association between d_{KL} and $d_{\mathbf{f},\mathbf{g}}$ either stronger or at least detectable.

(iii) By contrast, the sup-log-ratio route *does* certify.

Proposition 16 (Non-vacuity is capped). $\sum_l \text{Var}[z'_l - w'_l] \geq 2Md_{\text{SVD}}$ by trace $\leq$ nuclear norm, so bound $\geq \sqrt{2\lambda_{\max}d_{\text{SVD}}}$ and bound < 1 forces $d_{\mathbf{f},\mathbf{g}} < 1/(2\lambda_{\max}) \leq \frac{1}{2}$.

Whether that cap binds is empirical, and on CIFAR-10 it does not: $\lambda_{\max} = 1.44$ – 3.46 over $M = 2, 3, 5$ gives ceilings 0.347 – 0.145 against median $d_{\mathbf{f},\mathbf{g}}$ of $0.128, 0.065, 0.103$. So of the three obstructions only two are real; the third was a technique gap.

4.2 A diagnostic instead of a certificate

Four quantities, one forward pass, bounding in advance what a likelihood comparison can reveal. Tab. 2 works through a real pair. δ_{M+1} says whether KL can say anything; $\lambda_{\max}$ fixes the largest dissimilarity the method could ever rule out, and grows with M , so the certificate is a low-dimensional instrument; $\Delta/\psi_{\min}$ sets the order Cor. 13 would require; and α_{sym}^* says where on the Rényi scale the pair becomes distinguishable, being infinite exactly when the models are equivalent.

Table 2: The diagnostic applied to a pair of retrained CIFAR-10 models ($M = 2$). Four numbers, one forward pass, read before any similarity claim.

quantity	value	what it says
δ_{M+1}	1.1×10^{-15}	KL is informative only below $\epsilon \approx 10^{-31}$; observed $d_{\text{KL}} = 0.60$, so
λ_{max}	1.442	KL certifies nothing ceiling $1/(2\lambda_{\text{max}}) = 0.347$; pairs above this can never be certified
bound	0.931	below 1, so the pair is certified; the observed $d_{\text{f,g}}$ is well inside it
Δ/ψ_{min}	11.3	Cor. 13 would need $\alpha > 64$, far above the threshold near 1.35

Table 3: α_{sym}^* saturates in representation dimension. Margin floor 0.08, $d_{\text{f,g}} \geq 0.10$ required, 200 random geometries per M ; n survives both filters. Medians sit at 1.00: almost every non-equivalent pair has a critical order barely above 1.

M	2	3	4	6	8	12
n	144	199	192	182	177	183
best α_{sym}^*	1.34	1.31	1.20	1.26	1.18	1.22
median	1.004	1.002	1.001	1.001	1.002	1.002
$d_{\text{f,g}}$ at best	0.12	0.33	0.32	0.42	0.24	0.41

4.3 Open problem: how large is α_0 ?

Thm. 3 and Thm. 11 confine the controlling orders to $(1, \alpha_0]$, and Cor. 13 turns any bound on α_0 into a control modulus. What remains open is whether $\alpha_0 < \infty$ *uniformly*, without fixing a dissimilarity level.

All our searches place it near 1.3 (Tab. 3). Exhaustively over the 720 permutations at $k = 6$, order-breaking pairs reach 1.09 and reflections 1.11; a factorial over input concentration, angular regularity, norm profile and dimension tops out at 1.20; a targeted search over reflection-composed permutations (921 configurations) reaches 1.14. Two structural facts explain the ceiling. First, $\alpha_{\text{sym}}^* \rightarrow 1$ whenever an input approaches a decision boundary of one model while the other ranks the tied labels differently, and bisectors have codimension one. Second, at $M = 2$ the unembeddings carry a cyclic order, and an order-breaking permutation cannot be realised without a collision: the census shows $\alpha_{\text{sym}}^* = \infty$ exactly on the rotations, which are globally linear and have $d_{\text{f,g}} = 0$, while every order-breaking permutation is pinned near 1.05 however dissimilar its representations. The obstruction is topological, not metric — flat in the number of order violations and unmoved by non-uniform norms.

5 Experiments

Full protocols, sample sizes and negative controls are in App. E–F. Rank correlations carry 95% cluster bootstrap intervals over *seeds*: n seeds give $\binom{n}{2}$ dependent pairs, so pair-level resampling is anticonservative. No pair was dropped and no bound was violated anywhere.

Guards. The analysis reports rather than silently drops: the count of discarded pairs (zero everywhere), the per-pair argmax agreement (median 0.804 on CIFAR-10, minimum 0.607), and the assertion $\alpha_{\text{sym}}^* = \infty \Rightarrow d_{\text{f,g}} < 5/n$ implied by Thm. 11, whose tolerance is $5/n$ because the residual is $O(1/n)$ estimator bias rather than noise. These are released as a test suite; three substantive errors during this work were silent filters, and we recommend reading the guard lines before the results.

Synthetic: wider networks learn closer distributions and more dissimilar representations. 400 models ($c \in \{4, 6, 10, 18\}$ classes, widths 16–256, 20 seeds, 15k steps), 3,800 pairs. Mean training loss is flat in width ($0.0202 \rightarrow 0.0208$ at $c=4$), so width and fit quality are separated. In that regime mean d_{KL} *falls* with width ($5.06 \rightarrow 1.65$ at $c=10$) while mean $d_{\text{f,g}}$ *rises* ($0.436 \rightarrow 0.697$); loss-adjusted regression leaves the slope positive in all four class counts (+0.034 to +0.098). The effect is not an artefact of our invariance class: under m_{CCA} , the measure used in prior width studies, similarity falls monotonically with width in every class count ($0.741 \rightarrow 0.666$ at $c=4$; $0.639 \rightarrow 0.530$ at $c=6$), after applying the $\geq 90\%$ accuracy filter of Roeder et al. [2021]. Applying that filter removes 1,080 of 3,800 pairs but leaves the trend untouched where it is applied evenly: filtered means agree with the unfiltered ones to three decimals at $c=4, 6, 10$ (e.g. $0.4345 \rightarrow 0.6969$ against $0.436 \rightarrow 0.697$ at $c=10$). We rest the claim on $c=4$ and $c=6$, where the filter removes essentially nothing; the $c=18$ cells retain as few as three pairs and are uninformative (App. F). This is the paper’s thesis made visible on trained models at scale, and it is consistent with recent reports that convergence trends under CKA are confounded by width and depth [Gröger et al., 2026, Koepke et al., 2026, You et al., 2026]. We vary hidden width at fixed M and fixed dataset size, so the result speaks only to that slice.

Control curves. Tab. 5 reports the empirical modulus $\hat{h}(\epsilon) = \sup\{d_{\text{f,g}} : D \leq \epsilon\}$ — the direct counterpart of Def. 1 — at quantiles of each divergence. All divergences show substantial empirical control on *trained* models, with the higher orders falling further ($\hat{h}$ ratio 0.075 at $\alpha \geq 1.5$ against 0.113 for KL and 0.313 for total variation, the least tail-sensitive). The ordering matches the theory but the effect is weak: over 2,720 pairs surviving the accuracy filter, d_{KL} still tracks $d_{\text{f,g}}$ at $\rho = 0.602$ (95% cluster bootstrap

Table 4: Temperature sweep, CIFAR-10, 435 pairs from 30 seeds. Rescaling logits leaves $d_{f,g}$ invariant to machine precision, so only the confidence floor moves. Intervals are 95% cluster bootstrap over seeds. A paired bootstrap on the same pairs puts $\rho(T=16) - \rho(T=1)$ at $+0.109$ with interval $[+0.044, +0.202]$, so the rise is significant.

T	δ_3	mean d_{KL}	$d_{f,g}$	ρ [95% CI]
1	7.9×10^{-11}	0.601	0.1918	0.652 [0.37, 0.80]
2	8.5×10^{-6}	0.409	0.1918	0.689 [0.42, 0.83]
4	2.5×10^{-3}	0.289	0.1918	0.714 [0.46, 0.85]
8	3.5×10^{-2}	0.183	0.1918	0.736 [0.50, 0.86]
16	9.6×10^{-2}	0.096	0.1918	0.761 [0.54, 0.88]

over seeds $[0.506, 0.705]$, so the worst case our constructions realise is not generic under training. We regard this as delimiting the theory rather than undermining it.

CIFAR-10: a temperature intervention isolates the mechanism. Sixty ResNet18 [He et al., 2016] models on CIFAR-10 [Krizhevsky and Hinton, 2009], $M \in \{2, 3, 5, 8\}$, ten seeds each plus a smoothing arm, and a 30-seed arm at $M=2$ for the temperature study; 705 pairs in total. Rescaling logits by $1/T$ leaves $d_{f,g}$ *exactly* invariant, since d_{SVD} standardises its inputs, while δ_3 rises: this holds representational dissimilarity fixed and moves only the confidence floor. Two things are then facts rather than inferences (Tab. 4): δ_3 climbs nine orders of magnitude, and mean $d_{f,g}$ stays at 0.1918 with maximum change 0.00.

The rank correlation between d_{KL} and $d_{f,g}$ then rises monotonically across all five temperatures, $0.652 \rightarrow 0.761$. Because the same pairs appear at every T , the claim is about the *change* in ρ , and a paired cluster bootstrap over seeds gives $\rho(T=16) - \rho(T=1) = +0.109$ with 95% interval $[+0.044, +0.202]$ and $P(\text{increase}) = 0.9998$. Raising the confidence floor strengthens the association between likelihood agreement and representational agreement, with representational dissimilarity held exactly fixed — which is Cor. 15 in its stronger form.

We note that this required 30 seeds. At 10 the same experiment gave $+0.090$ with interval $[-0.049, +0.395]$: the effect was present but not establishable, because the binding resource is the number of independent training runs, not the number of temperatures. The dimensional comparison of Tab. 7 retains 10 seeds per dimension so that $M=2, 3, 5$ stay balanced.

Positive control. Every experiment above asks whether the theory *detects* a difference. Thm. 11 also makes the converse claim, which we test directly: applying a known invertible $\mathbf{A}$ to a trained model ($\mathbf{f} \mapsto \mathbf{A}\mathbf{f}$, $\mathbf{g}_0 \mapsto \mathbf{A}^{-\top}\mathbf{g}_0$) produces a pair that is $\sim_L$ -equivalent by construction. Across eight CIFAR-10 models with $\text{cond}(\mathbf{A}) \in [2.4, 5.1]$, all three predictions hold: $\alpha_{\text{sym}}^* = \infty$ in every case,

Table 5: Empirical modulus $\hat{h}(\epsilon)$ on 3,800 trained synthetic pairs, at quantiles of each divergence. Last column is $\hat{h}(q_{.05})/\hat{h}(q_{.5})$; smaller means the modulus falls faster as $\epsilon \rightarrow 0$, i.e. closer to control.

	$q_{.05}$	$q_{.25}$	$q_{.5}$	ratio
Total variation	0.259	0.750	0.827	0.313
Rényi 0.5	0.093	0.827	0.827	0.113
KL ($\alpha=1$)	0.093	0.633	0.827	0.113
Rényi 1.5	0.056	0.633	0.750	0.075
Rényi 2	0.072	0.633	0.750	0.096
Rényi ∞	0.072	0.673	0.949	0.075

Table 6: Mean m_{CCA} against hidden width, after the $\geq 90\%$ accuracy filter of Roeder et al. [2021] (315/400 models kept). m_{CCA} is a *similarity*, so a falling row is rising dissimilarity. The $c=18$ block is reported for completeness only: the filter removes models so unevenly there that its cells compare differently-selected subpopulations.

classes c	hidden width w				
	16	32	64	128	256
<i>mean m_{CCA}</i>					
4	0.741	0.689	0.692	0.681	0.666
6	0.639	0.597	0.562	0.533	0.530
10	0.733	0.559	0.545	0.459	0.437
18	0.833	0.625	—	0.490	0.451
<i>seed pairs retained</i>					
4	190	190	190	190	190
6	190	190	171	190	190
10	136	120	120	171	190
18	45	3	0	10	28

D_α vanishes to machine precision at $\alpha \in \{0.5, 1, 2, \infty\}$ ($\max |D_\alpha| \leq 1.3 \times 10^{-14}$), and $d_{f,g} = 1.000 \times 10^{-4}$ — *exactly* $1/n$ at $n = 10,000$ test inputs. That last value is worth noting: the residual on an equivalent pair is a deterministic estimator bias of $1/n$, not sampling noise, which is why our regression tolerance of $5/n$ carries a fivefold margin. The same holds at $M=8$, where $\text{cond}(\mathbf{A})$ reached 1.6×10^3 without affecting any of the three.

Negative controls. Two results we report because they failed. First, a pointwise trade-off holds between $\alpha^*(x)$ and the signal $\max_y(b_y - a_y)$ at fixed x ($r = -0.58$ over 531 geometries), but it does *not* survive averaging: over an input distribution the correlation falls to -0.16 , and concentrating the inputs moves α_{sym}^* and $d_{f,g}$ in the *same* direction ($1.013 \rightarrow 1.234$ against $0.665 \rightarrow 0.701$). The mechanism is Fig. 1(c): a minimum over inputs is pinned by the worst-margin input, whatever the representations do. Second, the label-smoothing arm on CIFAR-10 is confounded — smoothing costs accuracy ($0.737 \rightarrow 0.633$ at $s = 0.2$) and moves $d_{f,g}$ as well as δ_m (0.192, 0.153, 0.053, 0.143), and

Table 7: Certification and KL’s usefulness on CIFAR-10 across representation dimension; 435 pairs at $M=2$ (30 seeds) and 45 elsewhere (10 seeds), no violations anywhere. Both degrade with M by independent mechanisms: $\lambda_{\max}$ grows, tightening the ceiling, and δ_{M+1} reaches deeper into the tail. Certification reaches 0% at $M=8$, as predicted. Intervals are 95% cluster bootstrap over seeds.

	certification				KL’s usefulness			
	$M=2$	$M=3$	$M=5$	$M=8$	$M=2$	$M=3$	$M=5$	$M=8$
$\lambda_{\max}$	1.529	1.904	3.458	4.848	Spearman($d_{\text{KL}}, d_{\text{f,g}}$)			
ceiling $1/2\lambda_{\max}$	0.327	0.263	0.145	0.103	0.652	0.722	0.143	0.080
median $d_{\text{f,g}}$	0.113	0.065	0.103	0.093	$d_{\text{f,g}}$ as a fraction of the ceiling			
fraction of ceiling	0.35	0.25	0.71	0.90	δ_{M+1}			
median bound, naive	1.524	1.794	2.904	3.826	10^{-14}	10^{-14}	10^{-16}	10^{-22}
median bound, refined	0.895	0.764	1.226	1.368				
non-vacuous, refined	59%	64%	13%	0%				

the resulting correlations are non-monotone (0.725, 0.640, 0.338, 0.911). We draw no conclusion from it; the temperature design exists precisely to remove this confound.

Both failure modes track dimension, and the breakdown point was predicted. Tab. 7 sets the two side by side across $M = 2, 3, 5, 8$. They fall together by independent mechanisms — δ_{M+1} reaching deeper into the tail, $\lambda_{\max}$ growing and tightening the ceiling. From the first three dimensions we predicted in an earlier draft that certification would fail outright near $M \approx 7$; at $M=8$ it is 0%, and the KL- $d_{\text{f,g}}$ correlation is 0.080 ($p = 0.60$). Both predictions hold.

One does not. We expected the structural cap of Prop. 16 to bind first, and it does not: median $d_{\text{f,g}}$ is 0.093 against a ceiling of 0.103. The margin has closed steadily — $d_{\text{f,g}}$ is 35%, 25%, 71% and 90% of the ceiling at $M = 2, 3, 5, 8$ — but certification fails before the cap is reached, because the achieved bound (1.368) exceeds 1 while the cap still permits certification in principle. The obstruction at $M=8$ is therefore the constant, not the structure; the structural limit lies just beyond. Low-dimensional representations are the benign case for both halves of the theory, not the representative one. Crucially the four similarity measures agree *within* each dimension and weaken together across them: rank correlations with d_{KL} are $(+0.725, -0.768, -0.806, +0.806)$ at $M=2$ and $(+0.722, -0.805, -0.740, +0.799)$ at $M=3$ for $(d_{\text{f,g}}, m_{\text{CCA}}, \text{CKA}, \text{Procrustes})$, collapsing to $(+0.143, -0.116, -0.082, +0.067)$ at $M=5$. The dimensional degradation is therefore a property of the models, not of our invariance class. Pooling across dimensions — which mixes these populations — would report a misleadingly weak $+0.623$.

Why certification degrades with dimension. $\lambda_{\max}$ is the largest eigenvalue of $\text{Corr}(z_1)$, whose trace is M , so $\lambda_{\max} \in [1, M]$. Measured over $M = 2\text{--}32$ on constructed models it is linear in M with $R^2 = 0.993$, and

the slope equals the mean pairwise correlation $\bar{\rho}$ of the log-ratio coordinates (Fig. 2): for an equicorrelated matrix $\lambda_{\max} = 1 + (M - 1)\bar{\rho}$. The correlation arises structurally, since every coordinate $z_{1,l} = \log p(y_l|x) - \log p(y_0|x)$ shares the same pivot term. The slope is setting-dependent — 0.49 on constructed models and 0.64 on CIFAR-10 over $M = 2, 3, 5, 8$ ($R^2 = 0.97$) — so we report the formula rather than a universal constant. Either way the ceiling $1/(2\lambda_{\max})$ falls like $1/M$ and the method is a low-dimensional instrument.

Scope: language models. The family is motivated by autoregressive language modelling, where diversity is easily met ($|\mathcal{Y}| \sim 10^4$ against $M \sim 10^3$) but confidence is extreme — only a few next tokens are ever plausible. Our results predict the outcome there without running it. With $\lambda_{\max} = 1 + (M - 1)\bar{\rho}$ and $\bar{\rho} \in [0.49, 0.68]$ across the two settings we measure, the Prop. 16 ceiling at $M = 64$ lies between 0.011 and 0.016, far below any plausible $d_{\text{f,g}}$ between retrained models, while δ_{M+1} can only be worse than the 10^{-15} we measure at $M = 2$. Thm. 8 should therefore certify nothing at that scale and d_{KL} should carry no representational information — a falsifiable prediction rather than an evasion, and one our released code tests directly. The quantity to check is $\bar{\rho}$ in a tied-embedding transformer: if it falls outside $[0.49, 0.68]$, the formula rather than the conclusion is what needs revising.

A caveat on the worst case. Our constructions realise a worst case that training does not generically produce: on trained models KL retains a visible association with $d_{\text{f,g}}$ even where Thm. 10 guarantees nothing at all. The theory describes what *can* go wrong, and the diagnostic of Sec. 4.2 says when it is likely to.

6 Related Work

Similarity measures and their invariances. Comparing representations means choosing what to be invariant to, and the choice is contested: CCA [Hotelling, 1936] and its vari-

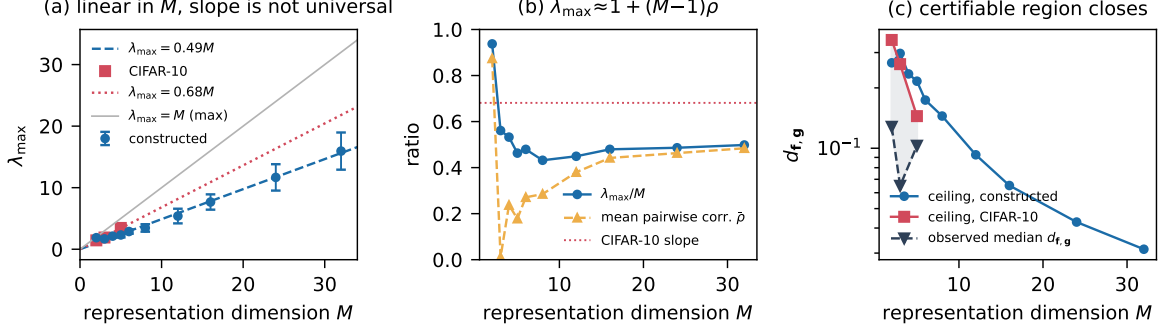


Figure 2: (a) $\lambda_{\max}$ is linear in M on constructed models ($R^2 = 0.993$ over $M = 2\text{--}32$), but the slope is not universal: 0.49 constructed against 0.68 on CIFAR-10, both well below the maximum $\lambda_{\max} = M$. (b) The slope is the mean pairwise correlation of the log-ratio coordinates: for an equicorrelated matrix $\lambda_{\max} = 1 + (M - 1)\bar{\rho}$, so $\lambda_{\max}/M \rightarrow \bar{\rho}$, and the two curves converge. The coordinates $z_{1,i}$ share a common pivot term, which induces the correlation. (c) The certifiable region $d_{f,g} < 1/(2\lambda_{\max})$ therefore closes as M grows; shading marks where observed CIFAR-10 dissimilarity still fits beneath it.

ants [Raghu et al., 2017, Morcos et al., 2018] admit any invertible linear map, CKA [Kornblith et al., 2019] only orthogonal transformations and isotropic scaling, shape metrics [Williams et al., 2021] the orthogonal case, while RSA [Kriegeskorte et al., 2008] compares similarity matrices. Klabunde et al. [2025] survey the field and observe that functional and representational measures need not correlate. Recent empirical work sharpens this: the scaling trends read as evidence for the Platonic Representation Hypothesis [Huh et al., 2024] are confounded by width and depth, with alignment largely vanishing under permutation-based null calibration [Gröger et al., 2026]; cross-modal convergence degrades as evaluation sets grow [Koepke et al., 2026]; and what survives appears to be neighbourhood structure rather than metric geometry [You et al., 2026]. Our width finding is consistent with this line, but differs in method: our invariance class is fixed by identifiability rather than chosen.

Functional similarity, and what it does not settle. A parallel line compares models by what their representations *support* rather than how they are shaped. Model stitching [Bansal et al., 2021] inserts a trained affine map between one network’s early layers and another’s late ones and asks whether performance survives, which tests whether representations are interchangeable for the task. Klabunde et al. [2025] document that functional and representational measures need not agree, and Sucholutsky et al. [2023] argue the field lacks a shared vocabulary for what alignment should mean. Our question is upstream of both: we ask what can be inferred about representations from the *output distribution alone*, without training a probe or a stitching layer. That matters because output distributions are the one thing always available — from an API, from a checkpoint, from a published log — and our answer is that the inference is weaker than it looks, in a way that depends on the

divergence and on how confident the models are.

Identifiability, exact and approximate. The softmax family is identifiable up to invertible linear maps under diversity, shown for conditional energy-based models [Khemakhem et al., 2020], the discriminative formulation [Roeder et al., 2021] and the multi-task setting [Lachapelle et al., 2023]. Marconato et al. [2025] lift diversity, characterising distribution-equivalent next-token predictors up to an extended-linear relation and obtaining an “all or none” dichotomy structurally analogous to our Thm. 11, though theirs concerns which properties transfer and ours when a divergence can detect non-equivalence. Liu et al. [2026] prove next-token representations are a linear transformation of latent posterior log-probabilities up to an $o(\epsilon)$ term. What this literature supplies is the relation $\sim_L$; what it does not supply is any statement about approximate agreement.

Robustness under approximate likelihood maximisation. Reizinger et al. [2024] introduce ϵ -non-identifiability and separately show cross-entropy can invert the data-generating process [Reizinger et al., 2025]. Buchholz and Schölkopf [2024] give the first robustness result for non-linear ICA under misspecification and note that not every identifiability result survives slight misspecification — the observation we make quantitative here. Nelson et al. [2026] distinguish statistical from structural near-identifiability. Related failures are known for correlated factors [Träuble et al., 2021] and independent mechanism analysis [Sliwa et al., 2022]. Closest to us, Nielsen et al. [2025] prove small KL does not guarantee representational similarity; we take that as the boundary case $\alpha = 1$, extend it to every $\alpha \leq 1$, and supply the missing other side.

Rényi order as a dial on worst-case ratios. The family and its monotonicity are classical [van Erven and Har-

remoës, 2014], and its extremes are exactly the regimes we contrast. Differential privacy turns on the same distinction: ϵ -DP is a max-divergence bound, and Rényi DP [Mironov, 2017] interpolates between it and weaker average-case guarantees. The trade-off recurs in variational inference, where the Rényi bound interpolates between mode-seeking and mass-covering [Li and Turner, 2016], motivates tail-adaptive alternatives under heavy-tailed density ratios [Wang et al., 2018] and admits monotonic schemes [Daudel et al., 2023]; and in PAC-Bayes, where Rényi bounds replace the KL term [Bégin et al., 2016]. In each case the order governs how much weight the rarest events receive. We show representational identifiability belongs in the same taxonomy, and locate its threshold.

7 Discussion and Conclusion

Limitations. (a) Two of three positive results remain out of reach: Thm. 10 needs $\epsilon < 1.7 \times 10^{-3}$ at best and Cor. 13 needs $\alpha \approx 64$ on our models against a threshold near 1.35. Thm. 8 does certify, but the $d_{f,g} < 1/(2\lambda_{\max})$ ceiling bounds its reach and the residual slack lies entirely in passing from log-ratio perturbations to coordinate correlations. Thm. 12 is valid on every trained pair we tested but exceeds the realised $d_{f,g}$ by two to three orders of magnitude, so it is a structural statement rather than a usable bound. (b) Effect sizes here are small relative to seed noise. The temperature effect is +0.109 and needed 30 seeds to separate from zero; the dimensional comparison rests on 10 seeds per dimension and its intervals are correspondingly wide. Conclusions from pair-level p -values in this setting would be badly anticonservative, since n seeds give $\binom{n}{2}$ dependent pairs. (c) The worst case is not the typical case: KL retains substantial empirical association with $d_{f,g}$ on trained models even where our bounds say nothing. (d) Control is proved for embeddings and unembeddings, not intermediate layers; $d_{f,g}$ has a specific invariance class we do not claim is the right one; diversity binds when classes are fewer than representation dimensions. (e) Our width result varies hidden width at fixed M and fixed dataset size, and the synthetic arm applies the accuracy filter of Roeder et al. [2021] only at analysis time. (f) We predict rather than measure behaviour on language models (Sec. 5).

In practice. Three recommendations, all cheap. *Report δ_{M+1} beside any KL-based similarity claim.* It bounds in advance what that comparison can reveal, and on a CIFAR-10 classifier it is 10^{-15} — which is to say nothing, however small the measured divergence. *Prefer orders $\alpha > 1$ to KL when comparing confident models.* No order at or below 1 controls (Cor. 4), and empirically KL’s association with $d_{f,g}$ collapses with representation dimension (Tab. 7). *Read $\lambda_{\max}$ before trusting a perturbation-style guarantee.* It fixes the largest dissimilarity the method could ever rule out, and grows with the representation dimension, so a certificate

that works at $M = 2$ may be structurally unavailable at $M = 16$. More broadly, matched loss is not evidence of matched representations, and the gap widens precisely as models become more confident and their representations wider — the regimes the field is moving toward.

Conclusion. The representational error is an unweighted spread of log-likelihood ratios; divergences weight labels by mass; the Rényi order measures the gap. Below $\alpha = 1$ nothing controls, above it every genuinely distinct pair is eventually visible, and the interval between is closed by a computable constant. The known failure of KL is the boundary case of this statement rather than an isolated counterexample. What that buys in practice is narrower than one would like: two of our guarantees cannot bite for reasons we can prove, and the third certifies a majority of retrained CIFAR-10 pairs only at low representation dimension. Where certification is out of reach, four quantities computable in one forward pass still bound what a likelihood comparison could reveal. Two findings emerged that we did not seek — wider networks learning closer distributions and further-apart representations, and both failure modes tightening with dimension — and the second says the regime where this theory is usable is not the regime the field is moving toward. Closing that gap is the natural continuation.

Impact Statement

This is foundational work on when representation comparisons are meaningful. The main societal relevance is that overclaiming representational agreement from likelihood agreement can mislead interpretability and distillation pipelines.

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A Notation

symbol	meaning
$\mathbf{f}, \mathbf{g}$	embedding $\mathcal{X} \rightarrow \mathbb{R}^M$ and unembedding $\mathcal{Y} \rightarrow \mathbb{R}^M$
$\mathbf{f}_0, \mathbf{g}_0$	displaced maps $\mathbf{f}(x) - \mathbf{f}(x_0), \mathbf{g}(y) - \mathbf{g}(y_0)$
$\mathbf{L}, \mathbf{N}$	matrices with columns $\mathbf{g}_0(y_i), \mathbf{f}_0(x_i), i = 1..M$
$\sim_L$	linear equivalence: $\mathbf{f} = \mathbf{A}\mathbf{f}', \mathbf{g}_0 = \mathbf{A}^{-\top}\mathbf{g}'_0$
$r_y(x)$	log-ratio $\log p(y x) - \log p'(y x)$
$R_y(x)$	logit difference $\mathbf{f}(x)^\top \mathbf{g}(y) - \mathbf{f}'(x)^\top \mathbf{g}'(y)$
a_y, b_y	margin gaps $s_{\hat{y}} - s_y$ and $s'_{\hat{y}} - s'_y$
$E_\alpha(y)$	tilted exponent $\alpha a_y + (1 - \alpha)b_y$
α_{sym}^*	$\min\{\alpha^*(p p'), \alpha^*(p' p)\}$
z_1, z_2	$\mathbf{L}^\top \mathbf{f}(x), \mathbf{L}'^\top \mathbf{f}'(x)$
Δ_l	$z_{1,l} - z_{2,l} = r_{y_l} - r_{y_0}$
s_l, s'_l	$\text{std}_x(z_{1,l}), \text{std}_x(z_{2,l})$
$\text{sd}_l, \tau_l, \psi_l$	$\text{std}(\Delta_l), s_l - s'_l , \min(s_l, s'_l)$
$\lambda_{\max}$	largest eigenvalue of $\text{Corr}(z_1); \leq M$
δ_m	$\min_x \min_{y \in \mathcal{Y}_m} \min\{p(y x), p'(y x)\}$
Δ	$\sup_{x,y} a_y(x)$

B Proofs for Section 3

B.1 Co-concentration (Lemma 2)

Along the norm-growth family the logits are $c\rho s_y$ with s_y fixed, so $p(y|x) = e^{-c\rho a_y}/Z_\rho$ with $Z_\rho = 1 + \sum_{y \neq \hat{y}} e^{-c\rho a_y}$ and $a_{\hat{y}} = 0$. Since $a_y > 0$ for $y \neq \hat{y}$, each summand vanishes and $Z_\rho \rightarrow 1$, giving $p(\hat{y}|x) \rightarrow 1$. The angle-matching assumption gives the same $\hat{y}$ and the same limit for p' , hence $\|p - p'\|_{\text{TV}} \rightarrow 0$. Meanwhile $|r_y| = c\rho|b_y - a_y| + O(1) = \Theta(\rho)$ whenever $a_y \neq b_y$. $\square$

B.2 Rate lemma and critical order (Theorem 3)

Set-up. For $\alpha \notin \{0, 1\}$, $D_\alpha = \frac{1}{\alpha-1} \log \sum_y p^\alpha p'^{1-\alpha}$. Substituting $p \propto e^{-c\rho a_y}, p' \propto e^{-c\rho b_y}$,

$$\sum_y p^\alpha p'^{1-\alpha} = Z_\rho^{-\alpha} Z_\rho'^{\alpha-1} \left[1 + \sum_{y \neq \hat{y}} e^{-c\rho E_\alpha(y)} \right], \quad E_\alpha(y) = b_y + \alpha(a_y - b_y). \quad (5)$$

(i) **Decay below threshold.** Expanding the three logarithms in (5) to first order,

$$\log \sum_y p^\alpha p'^{1-\alpha} = \sum_{y \neq \hat{y}} \left[e^{-c\rho E_\alpha(y)} - \alpha e^{-c\rho a_y} + (\alpha - 1)e^{-c\rho b_y} \right] + o(\cdot).$$

Group the exponents $\{E_\alpha(y)\} \cup \{a_y\} \cup \{b_y\}$ by value; at value v the total coefficient is $\#\{E_\alpha = v\} - \alpha \#\{a = v\} + (\alpha - 1)\#\{b = v\}$. A label with $a_y = b_y$ satisfies $E_\alpha(y) = a_y = b_y$ and contributes $1 - \alpha + (\alpha - 1) = 0$: such labels cancel exactly and must be excluded. The decay rate $\kappa(\alpha)$ is the smallest $v > 0$ with non-vanishing total coefficient, giving $D_\alpha = \Theta(e^{-c\rho \kappa(\alpha)})$.

(ii) **The case $\alpha = 1$.** At $\alpha = 1$ we have $E_1(y) = a_y$, so every first-order coefficient vanishes and the expansion above is uninformative. Computing directly,

$$d_{\text{KL}} = \sum_y p(\log p - \log p') = c\rho \sum_{y \neq \hat{y}} (b_y - a_y) e^{-c\rho a_y} / Z_\rho + \log(Z'_\rho / Z_\rho),$$

whose leading term is $\Theta(\rho e^{-c\rho \min_{\mathcal{P}} a_y})$ with $\mathcal{P} = \{y : a_y \neq b_y\}$. The polynomial prefactor is the signature of the first-order cancellation.

(iii) **Growth above threshold.** If $E_\alpha(y) < 0$ for some y the bracket in (5) diverges like $e^{c\rho \max_y (-E_\alpha(y))}$; dividing by $\alpha - 1$ and taking the logarithm gives growth that is *linear* in ρ with slope $\max_y (-E_\alpha(y))/(\alpha - 1)$. As $\alpha \rightarrow \infty$ this tends to $\max_y (b_y - a_y)$, which is D_∞/ρ .

The threshold. $E_\alpha(y)$ is affine in α with $E_0(y) = b_y > 0$. If $a_y \geq b_y$ it is non-decreasing and never negative. If $a_y < b_y$ it decreases and vanishes at $\alpha = b_y/(b_y - a_y)$. Hence $\alpha^* = \min_{y: b_y > a_y} b_y/(b_y - a_y)$, and since $0 < b_y - a_y < b_y$ we get $\alpha^* > 1$ unconditionally. $\square$

Numerical verification. At a single input ($k = 7, \rho \leq 160$) the bisected crossover matches (3) to 2.8×10^{-6} relative in the forward direction and 1.2×10^{-4} in the reverse; across 295 geometries the median relative error is 0.00%. Exponential rates match to $< 2.3\%$ and linear slopes to $< 0.1\%$. Omitting the $a_y = b_y$ exclusion inflates predicted rates by 40–70%. Code: `theory/perx.py, theory/rates.py`.

B.3 Uniform separation (Theorem 5)

Write $\mathbf{L} = \rho \mathbf{U}$, where the columns of $\mathbf{U}$ are the fixed displaced unembedding directions, and likewise $\mathbf{L}' = \rho \mathbf{U}'$. The measure d_{SVD} standardises each coordinate of its arguments to unit variance before forming the cross-covariance, so it is invariant to $z \mapsto cz$ for any $c > 0$. Hence

$$d_{\text{SVD}}(\mathbf{L}^\top \mathbf{f}, \mathbf{L}'^\top \mathbf{f}') = d_{\text{SVD}}(\mathbf{U}^\top \mathbf{f}, \mathbf{U}'^\top \mathbf{f}'),$$

in which ρ does not appear. Along the norm-growth family the embeddings $\mathbf{f}, \mathbf{f}'$ are fixed by construction, so the right-hand side is a constant c of the geometry alone. By Thm. 11, $c = 0$ if and only if the pair is $\sim_L$ -equivalent; the construction is not, so $c > 0$. Therefore $\max d_{\text{SVD}} \geq c > 0$ uniformly in ρ . $\square$

Numerical check. d_{SVD} is constant to four decimals across $\rho \in \{1, 5, 20, 80, 320\}$ at every geometry tested (0.2123, 0.4527, 0.5996 for permuted-cluster fractions $q = 0.25, 0.5, 0.75$ at $k = 8$). Code: `theory/verify_b3.py`.

Remark 17 (A bound we do not claim). One might hope for c in closed form, for instance $c \geq q$ for the permuted-cluster fraction q . That is false: across 28 configurations spanning $k \in \{4, 8, 12, 20\}$ and $q \in [0.1, 0.9]$ the inequality fails in 24, and none of $q, 1 - |1 - 2q|, 2q(1 - q)$ or $\min(2q, 1)$ is a valid lower bound. Measured d_{SVD} ranges over $[0.05, 0.86]$ and depends on the permutation structure rather than on q alone. Positivity is all the argument needs, and Thm. 11 supplies it.

C Proofs for Section 4

C.1 Spectral step (Proposition 9)

Hoffman–Wielandt gives $\sum_i (\sigma_i - \tilde{\sigma}_i)^2 \leq \|E\|_F^2$ for the singular values of $\Sigma_{z'z'}$ and $\Sigma_{z'w'}$. By Cauchy–Schwarz, $\frac{1}{M} \sum_i |\sigma_i - \tilde{\sigma}_i| \leq (\frac{1}{M} \sum_i (\sigma_i - \tilde{\sigma}_i)^2)^{1/2} \leq \|E\|_F / \sqrt{M}$. This replaces the per-singular-value Weyl bound used previously and saves a factor $\sqrt{M}$. $\square$

C.2 Perturbation bound (Theorem 8)

Write $\mu_l = \mathbb{E}[z_{1,l}]$. Since $z_{2,l} = z_{1,l} - \Delta_l$,

$$z'_l - w'_l = (z_{1,l} - \mu_l) \left(\frac{1}{s_l} - \frac{1}{s'_l} \right) + \frac{\Delta_l - \mathbb{E}\Delta_l}{s'_l}.$$

The first term has standard deviation $|s'_l - s_l|/s'_l = \tau_l/s'_l$ and the second sd_l/s'_l , so $\text{std}[z'_l - w'_l] \leq (\tau_l + \text{sd}_l)/\psi_l$.

For the spectral step, $\|E\|_F^2 = \sum_j \sum_k \text{Cov}[z'_k, \Delta'_j]^2$. The inner sum is the squared norm of the projection of Δ'_j onto $\text{span}\{z'_k\}$ and is therefore at most $\lambda_{\max} \text{Var}[\Delta'_j]$, rather than the $M \text{Var}[\Delta'_j]$ obtained by applying the covariance inequality entrywise — which would require all M^2 entries to be simultaneously perfectly correlated. Combining with Prop. 9,

$$d_{\text{SVD}} \leq \frac{\|E\|_F}{\sqrt{M}} \leq \sqrt{\frac{\lambda_{\max}}{M} \sum_l \text{Var}[z'_l - w'_l]} = \sqrt{\lambda_{\max}} Q.$$

The same argument on $\mathbf{N}^\top \mathbf{g}(y)$ bounds the unembedding term, and $d_{\mathbf{f}, \mathbf{g}}$ is their maximum. If $D_\infty \leq \epsilon$ then $|\Delta_l| \leq 2\epsilon$ pointwise, giving $\tau_l, \text{sd}_l \leq 2\epsilon$ and $Q \leq 4\epsilon/\psi_{\min}$. $\square$

Why the two refinements matter. Aggregating coordinates in root mean square rather than taking the worst gains 1.35 – $1.57\times$; keeping τ_l separate from sd_l gains a further factor, since the difference of standard deviations is much smaller than the standard deviation of the difference (median τ_l/sd_l is 0.25 – 0.53). Jointly they tighten the bound by 1.65 – $2.37\times$, and on CIFAR-10 by $1.67\times$: median bound $1.558 \rightarrow 0.931$, non-vacuity $0\% \rightarrow 58\%$, with $0/45$ violations.

C.3 Non-vacuity ceiling (Proposition 16)

In standardised coordinates $\text{Var}[z'_l - w'_l] = 2(1 - \rho_l)$ with $\rho_l = (\Sigma_{z'w'})_{ll}$, so $\sum_l \text{Var}[z'_l - w'_l] = 2M - 2\text{tr}(\Sigma_{z'w'})$. For any square matrix the trace is at most the nuclear norm, and $\sum_i \sigma_i = M(1 - d_{\text{SVD}})$, whence $\sum_l \text{Var}[z'_l - w'_l] \geq 2Md_{\text{SVD}}$. Substituting into the bound of Thm. 8 gives bound $\geq \sqrt{2\lambda_{\max}d_{\text{SVD}}}$, so bound < 1 forces $d_{\text{SVD}} < 1/(2\lambda_{\max}) \leq 1/2$. $\square$

Remark 18. The inequality $\sum_l \text{Var} \geq 2Md_{\text{SVD}}$ is identity-level, so the cap binds any argument routed through a Frobenius-norm perturbation estimate. Escaping it requires bounding the singular values of $\Sigma_{z'w'}$ directly. Empirically the spectral form $\sqrt{2\lambda_{\max}(1 - \bar{\rho})}$ sits within 11–17% of this floor, so essentially all residual looseness is in passing from log-ratio perturbations to the correlations ρ_l .

C.4 KL on Θ_δ (Theorem 10)

For $y \in \mathcal{Y}_m$ and $p, p' \geq \delta_m$ there, the mean value theorem gives $|r_y| = |\log(p_y/p'_y)| \leq |p_y - p'_y|/\delta_m \leq \|p - p'\|_1/\delta_m$. Pinsker gives $\|p - p'\|_1^2 \leq 2D_{\text{KL}}(p||p')$, so $\mathbb{E}_x[r_y^2] \leq 2\epsilon/\delta_m^2$ by Jensen. Then $\rho_2 \leq 2\max_y(\mathbb{E}_x r_y^2)^{1/2} \leq 2\sqrt{2\epsilon}/\delta_m$, and Thm. 8 concludes. Only labels in $\mathcal{Y}_{\text{LV}} \subseteq \mathcal{Y}_m$ enter z_1, z_2 , so the floor need only be taken over $\mathcal{Y}_m$ and no tail-mass term appears. $\square$

Proposition 19 (The floor cannot be lifted). *For any model, $\delta_{M+1} \leq 1/(M+1)$, so (10) is below 1 only for $\epsilon < 1/(64(M+1)^2)$; at $M = 2$ this is $\epsilon < 1.7 \times 10^{-3}$.*

Proof. The $(M+1)$ -th largest entry of a probability vector is at most $1/(M+1)$, since $M+1$ entries each exceeding it would sum to more than one. $\square$

C.5 Dichotomy (Theorem 11)

$\alpha^*(p||p') = \infty$ means no y has $b_y > a_y$, and $\alpha^*(p'||p) = \infty$ means no y has $a_y > b_y$; together $a_y(x) = b_y(x)$ for all x, y . Since $a_y - b_y = R_{\hat{y}} - R_y$, this says $R_y(x) = c(x)$ is independent of y , i.e. $\mathbf{f}(x)^\top \mathbf{g}(y) = \mathbf{f}'(x)^\top \mathbf{g}'(y) + c(x)$. The additive $c(x)$ cancels in the softmax normaliser, so $p_{\mathbf{f}, \mathbf{g}} = p_{\mathbf{f}', \mathbf{g}'}$, and linear identifiability gives $\sim_L$ -equivalence under diversity. The converse is immediate: equal distributions give $a_y = b_y$. $\square$

Regression test. Over 329 pairs, $\max |d_{\mathbf{f}, \mathbf{g}}| = 5.0 \times 10^{-4}$ in the ∞ class against $\min |d_{\mathbf{f}, \mathbf{g}}| = 6.5 \times 10^{-2}$ in the finite class. The residual decays as $O(1/n)$ in the sample size (2.0×10^{-3} at $n = 500$, 6.3×10^{-5} at $n = 16000$), the finite-sample bias of the cross-covariance estimator; the CI tolerance is therefore $5/n$.

C.6 Graded control (Theorem 12) and the ceiling (Corollary 13)

If $\alpha_{\text{sym}}^* \geq A$ then $b_y - a_y \leq b_y/A$ when $b_y > a_y$ and $a_y - b_y \leq a_y/A$ otherwise, so $|a_y - b_y| \leq \max(a_y, b_y)/A \leq \Delta/A$. Since $a_y - b_y = R_{\hat{y}} - R_y$, the map $y \mapsto R_y(x)$ lies within Δ/A of the constant $R_{\hat{y}}(x)$, so its spread is at most $2\Delta/A$ and $|\Delta_l| = |R_{y_l} - R_{y_0}| \leq 2\Delta/A$. Applying Thm. 8 gives $d_{\mathbf{f}, \mathbf{g}} \leq 4\sqrt{M}\Delta/(\psi_{\min}A)$. Inverting, for any $d_0 > 0$, $\alpha_0(d_0) = \sup\{\alpha_{\text{sym}}^* : d_{\mathbf{f}, \mathbf{g}} \geq d_0\} \leq 4\sqrt{M}\Delta/(\psi_{\min}d_0) < \infty$. $\square$

D Algorithms

Algorithm 3: Representational dissimilarity $d_{\mathbf{f}, \mathbf{g}}$

Input: embeddings $\mathbf{f}, \mathbf{f}'$ on N inputs; unembeddings $\mathbf{g}, \mathbf{g}'$; label subset with $|\mathcal{Y}_m| \geq M+1$; pivots x_0, y_0

Output: $d_{\mathbf{f}, \mathbf{g}} \in [0, 1]$, zero iff the pair is $\sim_L$ -equivalent

$\mathbf{L} \leftarrow [\mathbf{g}_0(y_1), \dots, \mathbf{g}_0(y_M)]$, $\mathbf{L}'$ likewise;

$\mathbf{N} \leftarrow [\mathbf{f}_0(x_1), \dots, \mathbf{f}_0(x_M)]$, $\mathbf{N}'$ likewise;

$z_1 \leftarrow \mathbf{L}^\top \mathbf{f}(x)$, $z_2 \leftarrow \mathbf{L}'^\top \mathbf{f}'(x)$, $w_1 \leftarrow \mathbf{N}^\top \mathbf{g}(y)$, $w_2 \leftarrow \mathbf{N}'^\top \mathbf{g}'(y)$;

foreach $(u, v) \in \{(z_1, z_2), (w_1, w_2)\}$ **do**

 standardise each coordinate to zero mean, unit variance;

$\Sigma \leftarrow \text{Cov}(u', v')$; $d \leftarrow 1 - \frac{1}{M} \sum_i \sigma_i(\Sigma)$;

end

return max of the two values;

E Experimental Details

Constructed models. Unembeddings on a circle of radius ρ (or quasi-uniform on S^{M-1} for $M \geq 3$), embeddings sampled so each lies angularly nearest its own label. The second model permutes unembedding *directions* while keeping

norms attached to label indices, which preserves the top logit exactly; both models are required to share the argmax, and the retained fraction is reported.

Synthetic. 20,000 points from a 2D Gaussian ($\sigma = 3$), c opposite-slice classes. Embedding net: three LeakyReLU layers $2 \rightarrow w \rightarrow w \rightarrow w \rightarrow 2$; unembedding: a free $c \times 2$ table. Adam, batch 128, 15,000 steps, 20 seeds per cell, 190 pairs per cell, 3,800 total.

CIFAR-10. ResNet18 embedding with an M -dimensional output; unembedding net of three 128-wide LeakyReLU layers followed by a linear map to $\mathbb{R}^M$. Adam, batch 32, 20,000 steps, 10 seeds. 45 pairs per dimension; 270 pairs across dimensions and smoothing levels.

Statistics. Rank correlations are reported with 95% cluster bootstrap intervals over *seeds* rather than pairs: n seeds give $\binom{n}{2}$ dependent pairs, so pair-level resampling is anticonservative. Point estimates elsewhere are medians over pairs; App. E gives the procedure.

Reproducibility guards. The analysis reports, never silently drops: the count of dropped pairs (must be zero or be explained), the per-pair argmax agreement, and the assertion $\alpha_{\text{sym}}^* = \infty \Rightarrow d_{\mathbf{f},\mathbf{g}} < 5/n$. Three substantive errors during this project were silent filters; the released CI suite guards acceptance rate, the $5/n$ rule, the rate lemma in both directions, and the proved constants.

F Additional Results

Tail localisation. Restricting both models to the reference model’s top- k support recovers 44%, 65% and 89% of the full KL at $k = 2, 3, 5$ for $M=2$, rising to 67%, 84% and 95% at $M=5$; the $k=1$ entry is identically zero by construction. So on a ten-class problem KL is spread over roughly half the labels rather than concentrated on the argmax — weaker tail localisation than one might expect.

Averaging over inputs. Decay remains exponential under averaging ($R^2 = 1.00$ against 0.91 for a power-law fit) but the rate is set by the minimum margin in the support, falling from 0.193 to 0.018 when clusters touch the decision boundary.

α_{sym}^* **across geometries.** Exhaustively over all 720 permutations at $k = 6$: rotations give $\alpha_{\text{sym}}^* = \infty$ and $d_{\mathbf{f},\mathbf{g}} = -0.000$; reflections 1.109; order-breaking 1.047. No permutation gives both a large critical order and large dissimilarity. Non-uniform norms do not change this, and the median is flat across 2–8 cyclic-order violations — the obstruction at $M = 2$ is topological rather than metric. Randomised search over $M \in \{2, \dots, 12\}$ reaches 1.34 with medians near 1.00.

Positive control. Every experiment above asks whether the theory *detects* a difference. Thm. 11 also makes the converse claim, which we test directly: applying a known invertible $\mathbf{A}$ to a trained model ($\mathbf{f} \mapsto \mathbf{A}\mathbf{f}$, $\mathbf{g}_0 \mapsto \mathbf{A}^{-\top}\mathbf{g}_0$) produces a pair that is $\sim_L$ -equivalent by construction. Across eight CIFAR-10 models with $\text{cond}(\mathbf{A}) \in [2.4, 5.1]$, all three predictions hold: $\alpha_{\text{sym}}^* = \infty$ in every case, D_α vanishes to machine precision at $\alpha \in \{0.5, 1, 2, \infty\}$ ($\max |D_\alpha| \leq 1.3 \times 10^{-14}$), and $d_{\mathbf{f},\mathbf{g}} = 1.000 \times 10^{-4}$ — *exactly* $1/n$ at $n = 10,000$ test inputs. That last value is worth noting: the residual on an equivalent pair is a deterministic estimator bias of $1/n$, not sampling noise, which is why our regression tolerance of $5/n$ carries a fivefold margin. The same holds at $M=8$, where $\text{cond}(\mathbf{A})$ reached 1.6×10^3 without affecting any of the three.

Negative controls. A pointwise trade-off between $\alpha^*(x)$ and the signal $\max_y (b_y - a_y)$ holds at fixed x ($r = -0.58$ over 531 geometries) but does *not* survive averaging: over an input distribution the correlation falls to -0.16 , and input concentration moves α_{sym}^* and $d_{\mathbf{f},\mathbf{g}}$ in the same direction. The label-smoothing arm on CIFAR-10 is confounded, since smoothing costs accuracy ($0.737 \rightarrow 0.633$) and moves $d_{\mathbf{f},\mathbf{g}}$; its correlations are non-monotone and we draw no conclusion from them.